Module Summary Report

AI Programming Foundations Project
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Dataset: Iris Flower Measurements (UCI Machine Learning Repository)

1. Overview

This project implements a complete, reproducible data science workflow applied to the Iris flower dataset, a benchmark dataset widely used in pattern recognition research. The workflow covers data ingestion, cleaning, exploratory analysis, and visualization, and is designed to serve as a reusable foundation for subsequent machine learning, deep learning, and agentic AI projects in this Nanodegree program. The dataset is available at: https://archive.ics.uci.edu/dataset/53/iris.

2. Dataset Description

The Iris dataset, originally introduced by statistician Ronald A. Fisher in 1936, contains 150 observations of iris flowers drawn from three species: *Iris setosa*, *Iris versicolor*, and *Iris virginica* (Fisher, 1936). The dataset comprises 5 columns: four continuous numeric features — sepal length, sepal width, petal length, and petal width (all measured in centimetres) — and one categorical target column, species. Each species is represented by exactly 50 observations, making the dataset perfectly balanced. The key variables focused on in this analysis were petal length and petal width, as these dimensions showed the strongest between-species separation and the highest inter-feature correlation (Pearson $r = 0.96$). Sepal width was identified as carrying largely independent information from the other three features and was treated as a secondary variable.

3. Workflow Description

The workflow was structured into five sequential stages, following best practices for reproducible data science (Danchev, 2022):

Ingestion. The dataset was loaded from a local CSV file using Pandas' `pd.read_csv()` function. The first five rows were displayed with `df.head()` and column types were verified with `df.info()` to confirm successful loading.

Cleaning. Two modular cleaning functions were defined and applied: `drop_duplicate_rows()`, which removes exact duplicate records to prevent inflated counts and biased statistics; and `fill_missing_numeric()`, which imputes missing values in numeric columns using the column median, a method that is robust to outliers. Both functions include docstrings and print diagnostic messages, satisfying reproducibility requirements.

Exploratory Analysis. A single EDA function, `exploratory_analysis()`, was defined to compute and display: (a) overall descriptive statistics, (b) per-species mean values, and (c) a Pearson correlation matrix for all numeric features. This function can be reused on any tabular dataset with minimal modification.

Visualizations. Three labeled plots were produced using Matplotlib and Seaborn: a stacked histogram of petal length by species (Figure 1), a scatter plot of sepal length versus petal length coloured by species (Figure 2), and a Pearson correlation heatmap (Figure 3). Each plot was saved as a high-resolution PNG file.

Summary. A concluding Markdown section in the notebook synthesises findings, identifies limitations, and notes unexpected observations.

4. Key Decisions and Assumptions

Cleaning choices. Median imputation was chosen over mean imputation for missing value handling because the median is more robust to skewed distributions and outliers — a consideration relevant to biological measurement data where extreme values may arise from instrument error rather than true variation. This reflects the principle that data-handling decisions should minimise the introduction of systematic bias (Danchev, 2022). Duplicate removal was applied with a keep-first policy, which is appropriate when duplicates are assumed to arise from data entry redundancy rather than legitimate repeated observations.

Focus of EDA. Petal features were prioritised in the exploratory analysis because the correlation matrix revealed that petal length and petal width have higher discriminatory power across species than sepal measurements. Grouped means confirmed that petal dimensions separate *I. setosa* cleanly from the other two species.

Visualization design. Figure 1 (histogram) was designed to show distributional overlap, revealing that *setosa* is fully separable by petal length alone. Figure 2 (scatter) was chosen to visualise the joint relationship between two features and show class clustering. Figure 3 (heatmap) was included to give a compact, quantitative view of all pairwise correlations, directly informing future feature selection for ML models.

5. Results and Interpretation

The most important pattern identified in this analysis is the clear separability of *Iris setosa* from the other two species based on petal dimensions (Figure 1). Setosa petals range from approximately 1 to 2 cm in length, while versicolor and virginica occupy higher ranges with partial overlap between 4 and 5 cm. This suggests that a simple threshold classifier on petal length alone could achieve high accuracy for setosa detection.

Figure 2 reveals a strong positive linear relationship between sepal length and petal length across all species (overall Pearson r = 0.87), with species clusters visible as distinct regions in the scatter plot. The correlation heatmap (Figure 3) confirms that petal length and petal width are highly redundant (r = 0.96), which has practical implications: including both features in a future model may add little information while increasing complexity.

Grouped mean values showed that *I. virginica* has the largest petals on average (petal length mean = 5.55 cm), followed by versicolor (4.26 cm) and setosa (1.46 cm). An unexpected finding was that setosa has the largest average sepal width (3.43 cm) despite having the smallest petals overall, suggesting that sepal width is not simply a proxy for overall flower size.

6. Responsible Practice: Bias and Data Quality

Several data-handling decisions in this workflow carry potential for introducing bias if applied carelessly to other datasets. First, median imputation assumes that missingness is random (Missing Completely at Random, MCAR). If data is missing in a structured way — for example, if measurements are systematically absent for one species due to collection difficulties — imputing with the overall column median would draw that species' values toward the population central tendency, reducing between-species variance and potentially making classification harder.

Second, dropping duplicates without investigation assumes they arise from data entry errors. In datasets where duplicate records reflect repeated measurements of the same subject, removal would reduce sample size unevenly across groups if some subjects were measured more often than others. To reduce this risk in future projects, duplicates should be inspected before removal rather than dropped blindly.

Third, the Iris dataset itself is a controlled laboratory sample collected by a single researcher in 1936 (Fisher, 1936). As such, it does not represent the full genetic and environmental diversity of the three species. Models trained on it may fail to generalise to specimens collected in different geographic regions, seasons, or conditions.

7. Reproducibility

To ensure that any collaborator or reviewer can reproduce this work exactly, the following measures were applied, consistent with open science best practices (Danchev, 2022):

A `requirements.txt` file was generated using `pip freeze` and included in the repository. This file pins exact package versions for all dependencies (NumPy, Pandas, Matplotlib, Seaborn, etc.), allowing any user to recreate the environment with `pip install -r requirements.txt`.

The project is version-controlled using Git. The repository includes multiple commits marking progress at each workflow stage (setup, ingestion, cleaning, EDA, visualizations, documentation) and at least one additional branch beyond main, demonstrating professional branching practices. The full commit history provides a transparent audit trail of how the project evolved.

All random elements were avoided in this workflow (no sampling, no train/test splits), so no random seed is required. The notebook is designed to run top-to-bottom without errors when executed with **Kernel** → **Restart & Run All**.

8. Sources and Citations

The following sources were cited in this report. In-text citations use (Author, Year) format; the reference list follows APA 7th edition.

References

Danchev, V. (2022). Reproducible Data Science with Python: An Open Learning Resource. *Journal of Open Source Education*, *5*(56), 156. https://doi.org/10.21105/jose.00156

Fisher, R. A. (1936). The use of multiple measurements in taxonomic problems. *Annals of Eugenics*, *7*(2), 179–188. https://doi.org/10.1111/j.1469-1809.1936.tb02137.x

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